

Generation Interconnection System Impact Study Report

For Queue Project Q22-2191

State Street 100 kV

November 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2191. The Interconnection Customer proposes a 190.5 MW solar photovoltaic facility interconnecting at the State Street 100 kV substation in the Indiana area. The request was received on September 27, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 685 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,109,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Barret Substation / Roseau County 100kV line
Contingency: Monte Del Estado 6303 / Hrper 100kV
Loading: 160.0% (Rating: 234 MVA)
DFAX: 5.4%
Pre-project loading: 155.6%
2. Facility: State St / W Olympia 100kV Ckt 1
Contingency: State Street / West Olympia 100kV
Loading: 156.0% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 149.4%
3. Facility: W Olympia / State Street 100kV Line 2
Contingency: West Olympia Substation / State St 100kV
Loading: 156.0% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 149.4%
4. Facility: West Olympia Substation / State Street 100kV
Contingency: State Street / W Olympia 100kV Circuit 1
Loading: 156.0% (Rating: 167 MVA)
DFAX: 5.8%
Pre-project loading: 149.4%
5. Facility: Hassen Cassel / Darlington Rd 100kV line
Contingency: Darlington Rd / Wayne Trace 100kV
Loading: 143.1% (Rating: 189 MVA)
DFAX: 5.3%
Pre-project loading: 137.8%
6. Facility: Muldoon Mill / Hassen Cassel 100kV
Contingency: Wayne Trace / Darlington Rd 100kV L
Loading: 139.3% (Rating: 204 MVA)

DFAX: 5.3%

Pre-project loading: 134.4%

7. Facility: Muldoon Mill / Hillrest 100kV

Contingency: Wayne Trace / Darlington Rd 100kV Line

Loading: 111.3% (Rating: 435 MVA)

DFAX: 8.4%

Pre-project loading: 107.7%

8. Facility: Monroeville / Woodburn 100kV

Contingency: Richland / Dekalb 500kV

Loading: 107.7% (Rating: 277 MVA)

DFAX: 5.1%

Pre-project loading: 104.2%

9. Facility: Sage / Woodburn 100kV

Contingency: Dekalb / Richland 500kV

Loading: 106.9% (Rating: 291 MVA)

DFAX: 5.1%

Pre-project loading: 103.5%

10. Facility: Sage / Monte Del Estado 6303 100kV L

Contingency: Richland / Dekalb 500kV

Loading: 105.9% (Rating: 311 MVA)

DFAX: 5.1%

Pre-project loading: 102.7%

11. Facility: Monte Del Estado 6303 / Harper 100kV

Contingency: Barret / Roseau County 100kV

Loading: 105.2% (Rating: 423 MVA)

DFAX: 6.8%

Pre-project loading: 102.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barret / Roseau County 100kV Line	Rebuild line with higher-capacity conductor	\$7,277,000	38% (\$2,784,000)
State St / W Olympia 100kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,660,000	100% (\$1,660,000)
State St Substation / West Olympia 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,601,000	100% (\$1,601,000)
State Street / West Olympia 100kV Ckt 3	Rebuild line with higher-capacity conductor	\$1,817,000	100% (\$1,817,000)
Hassen Cassel / Darlington Rd 100kV line	Rebuild line with higher-capacity conductor	\$1,690,000	39% (\$653,000)
Muldoon Mill / Hassen Cassel 100kV	Rebuild line with higher-capacity conductor	\$3,048,000	39% (\$1,178,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muldoon Mill / Hillcrest 100kV	Rebuild line with higher-capacity conductor	\$3,354,000	5% (\$167,000)
Monroeville / Woodburn 100kV	Reconductor line	\$3,402,000	40% (\$1,361,000)
Sage / Woodburn Substation 100kV line	Reconductor line	\$1,380,000	40% (\$552,000)
Monte Del Estado 6303 / Sage 100kV	Reconductor line	\$532,000	40% (\$213,000)
Monte Del Estado 6303 / Harper 100kV	Reconductor line	\$2,049,000	6% (\$123,000)
	Total allocated to project		\$12,109,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.